5G MAG The Media Connectivity Association



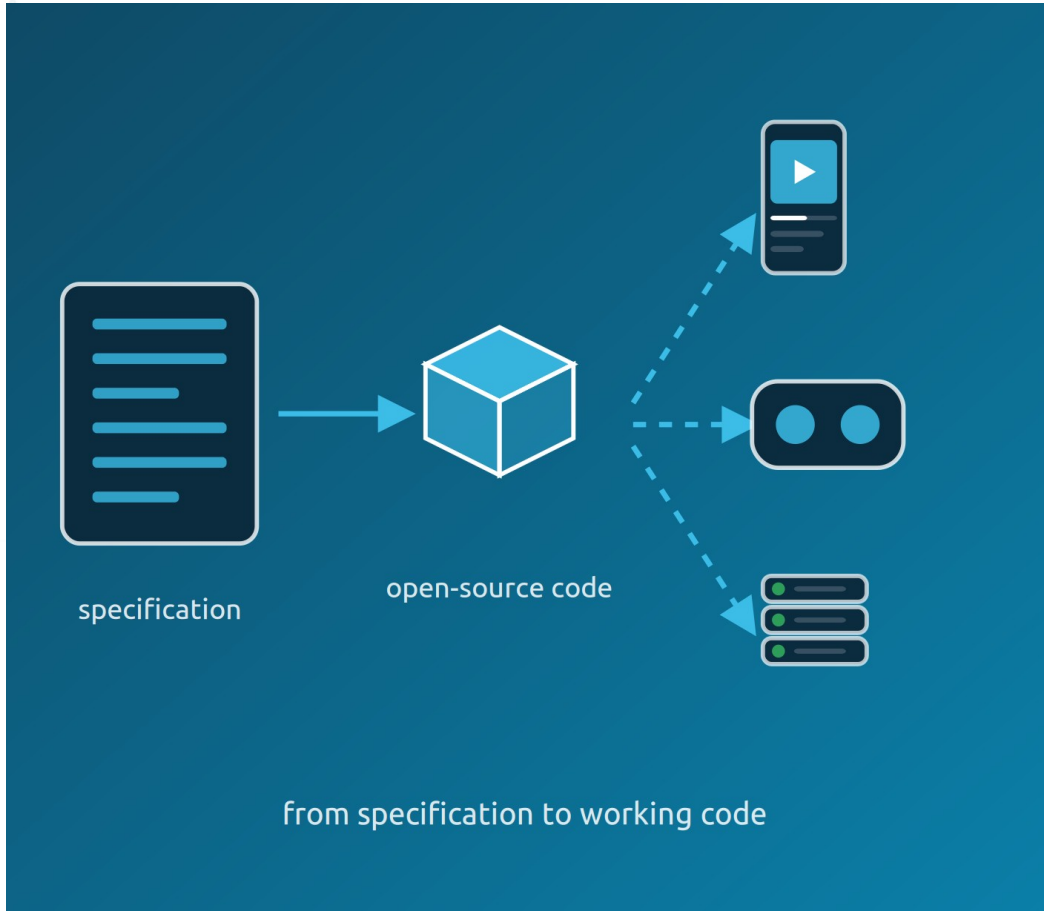

# Learn more about the Media Connectivity Software Accelerator

Open-source tools, validators and testbeds turning specs into working code

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# Specs into implementations, via open source

Open-source tools, validators and testbeds turning specs into working code




## Standards validation

Interop, early prototyping and testing.

## Ready-to-use code

For production environments and as benchmark.

## Shared effort

Shared development effort avoids duplication across companies.

## Transparency and trust

Collaborative enhancements.

# Media Connectivity Software Accelerator

Early feedback and validation

Software projects & Dev community

No duplication and fast deployment

Open by design and IPR-friendly

**Reference Tools**

Open-source implementations of the specifications.

**Testbeds and Evaluation Frameworks**

Environments to measure or evaluate something.

**Deployment Kits**

Packages, images and apps to put into a product. Coming soon.

# Enabling open-source software from the start of every project

| | TODAY: OPEN SOURCE AFTER THE FREEZE | WITH OPEN SOURCE FROM THE START |
|---|---|---|
| **Evaluation** | **Duplicated prototypes.** Each organization builds its own to evaluate a draft; results differ and feedback is slow. | **One shared baseline.** Drafts are validated against running code before the freeze, while gaps are cheap to fix. |
| **Interoperability** | **Late interop bugs.** Testing starts from scratch at plugfests; errors, some in the specification, surface late. | **Faster plugfests.** Implementers start from the same code: shorter preparation, faster root-cause analysis. |
| **Deployment** | **Slow deployment.** Years can pass from a frozen specification to production-ready code; early adopters carry the cost alone. | **Earlier deployment.** End-to-end demos and trials run before commercial products, and feed evidence back into the standard. |

1 Plan it at launch → 2 Engage implementers during drafting → 3 Make the value tangible

***"Open-source software and evaluation platform: home and scope identified at launch."***



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# Collaboration in technology development

Technology development and deployment, with and for the developers

## Supporting productization

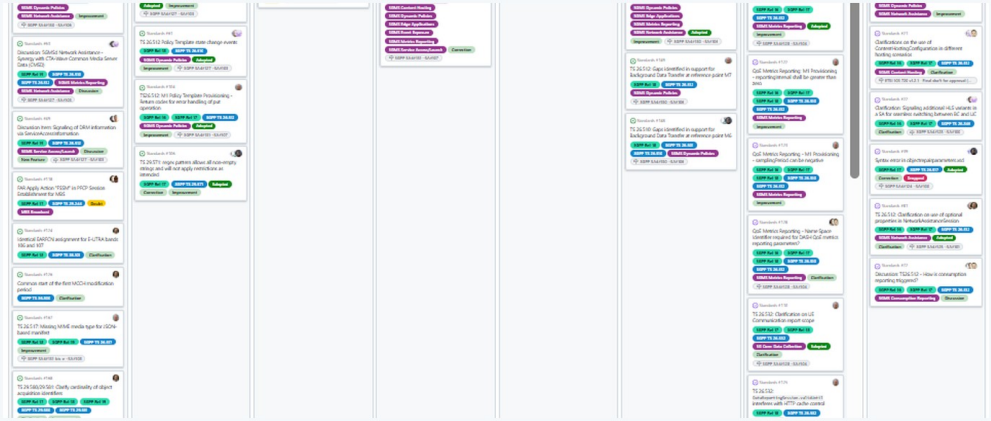
Prototypes, demos, trials, testbeds, …

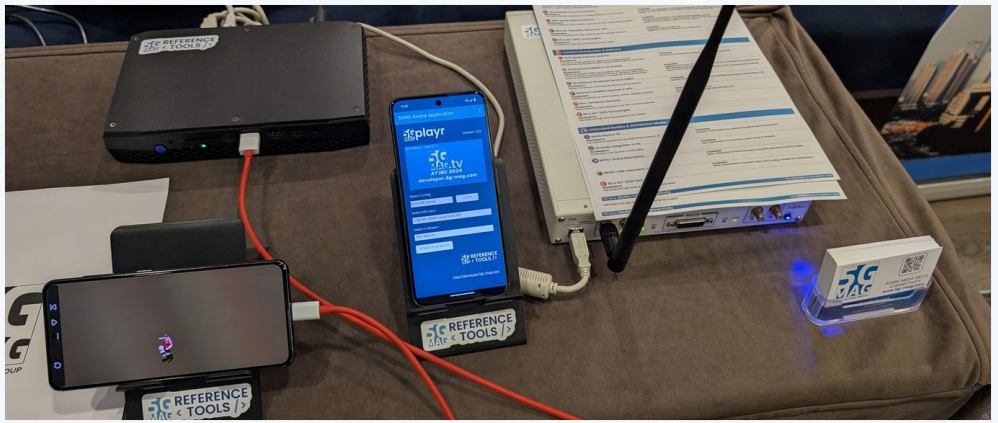
## Standardization feedback loop

Requirements and implementation feedback to standards development organizations (standards.5g-mag.com).

## Open to contributions

Sponsored by 5G-MAG. No requirement to be a 5G-MAG member.








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# 13 reference tools across 5 technology areas

## Content Delivery and Streaming

- 5G Media Streaming (5GMS) Architecture
- UE Data Collection, Reporting and Event Exposure
- DVB-I over 5G Systems
- Content Delivery Protocols and Tools

## Immersive Media

- XR/3D Scenes with MPEG-I Scene Description
- MPEG V3C Immersive Platform
- Dynamic Mesh Coding with MPEG V-DMC
- Conversational Avatars with MPEG ARF

## Broadcast and Multicast

- 5G Broadcast for TV and Radio Services
- 5G Broadcast for Emergency Alerts
- 5G Multicast Broadcast Services (MBS)

## Real-Time Communications (RTC)

- Real-time Media Communication (RTC) Architecture

## Connected Media Production

- CAMARA Connectivity Quality Management APIs

# Testbeds, evaluation frameworks and validators

Environments to measure or evaluate something, and tools that check conformance

## TESTBEDS

### 6G AI Traffic Characterization

AI traffic profiling and 5G-to-6G migration testbed.

Testbeds and evaluation frameworks ›

## EVALUATION FRAMEWORKS

### AI/ML Evaluation Framework

Framework for evaluating AI/ML solutions in mobile media services.

### Beyond 2D Evaluation Framework

Test and evaluation framework for immersive video quality assessment.

## VALIDATORS

### VOPS Conformance Validator

Conformance validator for the video operation points of 3GPP TS 26.265.

Validators ›

# Developer Exchanges and Tutorials

Talks by the developers, one per project, with their authors; step-by-step tutorials are on each tool's page




More information at **5g-mag.com/developer/exchanges**

# Community, maintenance and support

## OPEN AND ACCESSIBLE TO ALL

### Working group WG DEV

The members' workgroup for development and ecosystem, with weekly calls for 5G-MAG developers.

### Friday's Public Call

Regular public calls for all: the last Friday of the month, 13:00 CET/CEST.

### Slack

Fast communication with developers: the 5G-MAG Dev Community on Slack.

### Mailing list

News and announcements, with releases and news, on Google Groups.

## MAINTENANCE AND SUPPORT



### Coordination, testing and support

By Daniel Silhavy, Nils Duval and the 5G-MAG Project Office, in coordination with the contributors.

### Support for contributors

Guidelines and documentation; issue tracking, PR processing, roadmaps; dashboard, on-boarding.

### Integrity and deployability

GitHub Teams, 3GPP SA4 Standards Team; write and early access; CI/CD, Dockerization; REST API collections.

### Resources and visibility

GitHub resources, documentation, tutorials; promotion, visibility, attribution; additional resources.

# A note on the Open-Source Software Licenses

Protecting IPR. Enabling industry collaboration.

## What is the License Model?

1. The 5G-MAG Public License v1.0 is a modified version of Apache 2.0 which allows:
   - **Free Non-Commercial Use:** use, copy, modify and distribute for study, academic research, testing and validation.
   - **Commercial Exploitation:** use the software in commercial services, trials and live deployments, but:
     - **FRAND Patent Licensing:** essential contributor patents are made available under **FRAND** (Fair, Reasonable, and Non-Discriminatory) terms.
2. Repositories with software derived from other licenses keep those licenses.

## How to Contribute?




1. Sign an Individual or Corporate Contributor License Agreement (CLA).
2. Anybody can contribute: no 5G-MAG membership needed.
3. Members get priority support from the Project Office, and define where resources go.

# Growing community of developers

Organizations contributing to the Software Accelerator. We remain fully open and welcome more contributors joining efforts.

BBC

Big Blue Marble

Bitstem

Dolby Laboratories

EBU

Ericsson

Fraunhofer FOKUS

Huawei

idTOLU

InterDigital

iTEAM-UPV

Motion Spell

Nokia

Philips

phine.tech

Qualcomm

Resillion

Sofia Digital

# Software Accelerator in numbers

**19**

projects across media and connectivity

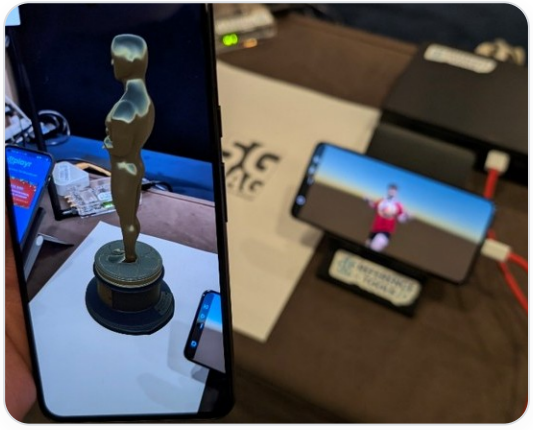


**68**

project repositories, public and early access

**+20,000**

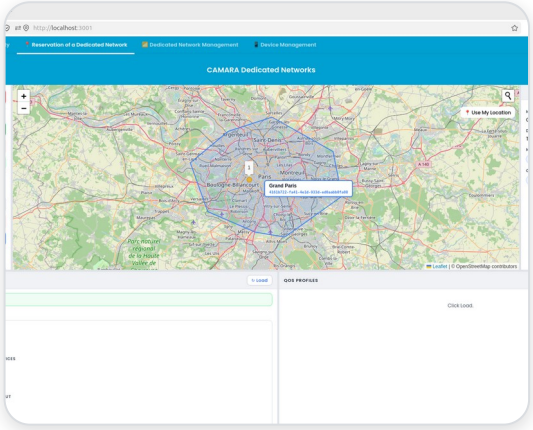
clones: unique cloners per day






**Monthly**

Dev Public Call and rolling releases



JOIN THE EFFORT

# Get involved

Ways to work with us, member or not

### Bring your topic or project

Don’t see your topic here? Bring it, or your project, as a member.

### Early Access

Request access to Reference Tools and Testbeds still in development.

### Contribute code

No membership needed: sign the CLA and open a pull request on GitHub.

### Join the developers

The Dev Community on Slack and the Reference Tools mailing list.

### Stay informed

Email updates, the 5G-MAG Magazine and The Handshake podcast.

### Follow us

News on LinkedIn; talks and demos on YouTube.

The Media Connectivity Association

# Join the Effort

Help shape what connected media looks like next and scale your projects with the industry.

GitHub

5g-mag.com

**Become a member →**

www.5g-mag.com/membership